

A Theory of Economic Slack

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2	PREVALENCE, CYCLICALITY, AND SOCIAL COST OF SLACK	3
2.1	Definition of economic slack	4
2.2	Prevalence of slack on the labor market	5
2.3	Prevalence of slack on the product market	10
2.4	Coexistence of sellers and buyers	13
2.5	Prevalence of long-term relationships	21
2.6	Cyclicalit y of slack	23
2.7	Social cost of slack	35
2.8	Summary	40
	BIBLIOGRAPHY	41

CHAPTER 2.

Prevalence, cyclical, and social cost of slack

Before we start building our slackish model of business cycles, it is critical to look at the data to get a good sense of what we want to model. As Kuhn (1957) explains, one of the three main qualities of a good model is to be descriptive—that is, to describe well the phenomena of interest. We therefore need to look at data from the real world to assess exactly how much slack there is out there, how slack behaves over time, and how costly slack might be. Throughout the book, we will incorporate these facts into the model—use them to justify model assumptions, calibrate model parameters, and discipline model mechanisms.

Everyone knows that during severe recessions such as the Great Depression or the Great Recession, a vast number of workers become unemployed. So, to understand these events, we must have a model that features unemployment in bad times.

What we will come to realize is that unemployment and slack are much more prevalent than one might think. It is not true that slack only occurs on the labor market during deep downturns: slack is a pervasive feature of modern economies, present in most markets at all times and in varying amounts. Therefore, a good macroeconomic model should feature slack in all markets, as well as fluctuations in slack. However, because slack is not presently a core concept in business cycle research, it is not systematically measured. Hence, all measures of slack reported in this chapter are imperfect in some ways; what matters is that they paint a consistent pattern across markets and over time.

The evidence assembled in this chapter draws on data from the United States. I make this choice for two reasons. First, the models in this book are designed to describe the US

economy, which is the benchmark economy in macroeconomics, and of course the largest global economy. International data are not directly applicable to the US economy because market-level and macroeconomic outcomes are shaped by country-specific regulations, institutions, and customs.

Second, the United States is the most demanding testing ground for a theory of slack. The United States is widely seen as the paragon of competition and free markets: the economy that should come closest to the Walrasian ideal in which slack does not exist. Yet, as we will see throughout this chapter, most US markets are actually riddled with slack—idle or unemployed workers, unsold goods, empty seats. If slack is pervasive even in the economy least hospitable to it, it is bound to be at least as pervasive elsewhere. In that way, the US focus sets the highest possible bar for the evidence.

Focusing on one country brings discipline to the data collection and model construction, but it certainly does not imply that the theory developed in this book does not apply to other countries. Chapter 21 shows that many patterns of slack observed in the US economy are also found, often more visibly, in other developed and developing economies. Slack therefore appears to be a universal feature of market economies.

2.1. Definition of economic slack

The concept of economic slack describes situations in which goods or services are available for sale but trades do not occur. When the goods or services are already produced, their costs of production are sunk, so the trades would typically generate a positive surplus. Sometimes the goods and services are not yet produced, but they are available to be produced at minimal marginal costs, so the main idea behind the concept of slack remains: trades that would generate value for sellers and buyers do not materialize.

There is no room for slack in the Walrasian model. In that model, anyone can sell anything at the market price. It is never the case that somebody would like to sell something at the market price but cannot find a buyer. Moreover, the marginal seller and buyer on a Walrasian market are indifferent between trading or not trading. Their marginal cost or marginal value for the good equals the market price, so it does not matter to them whether they trade or not. By contrast, in a world with slack, sellers are positively happy when they are able to trade, and they are distinctly hurt when they cannot trade. The same is true for buyers: they are positively happy when they are able to find the good that they are looking for.

The most famous example of slack is unemployment on the labor market. There are workers who are available and willing to work, and there are firms who are looking for workers, but the job seekers are unable to find jobs. As a result, the workers remain unproductive and their labor services are wasted.

But we also find economic slack in many other markets, not only in the labor market.

For instance, in the market for services, we very often see slack. A hairdresser might be idle if no customers have come to the hair salon to get a haircut. The hairdresser is at work since the salon must remain open to attract customers; but they are idle at work until a customer arrives. We see this too in restaurants and cafes, where not all tables are always full and there are sometimes more empty seats than at other times. Depending on the number of customers sitting in their cafes and restaurant, waiters and cooks are more or less busy. Slack is also observed in hotels and airplanes, which are not always full, with hotel rooms and airplane seats sometimes remaining empty.

Slack still exists in other markets, for example in the market for goods. There, some goods are produced but cannot be sold and are thrown away: for instance food that nobody buys and that expires. But it's not just perishable goods. We also find economic slack in durable goods, such as clothes and appliances that remain unsold. Although durable goods don't go bad immediately, they depreciate over time, so stores would later have to sell them at a lower price and, eventually, discard them.

We can also see slack within firms. People employed by firms do not work at full capacity at all times; they are more or less utilized depending on the demands on their time. This sort of slack is apparent in all types of jobs. Manufacturing workers might be forced to work shorter hours if production lines are temporarily shut down due to low demand, and sometimes they might be forced to work overtime. In consulting firms, consultants are more or less busy. When business is booming, consultants work long hours on projects. But sometimes, when business is slower, consultants stay home because there isn't enough business available for them to work. So slack also affects people who are employed within firms when firms do not have business.

2.2. Prevalence of slack on the labor market

We start by documenting the prevalence of the most well-known form of slack, and the most socially costly: the unemployment of workers on the labor market.

2.2.1. Definition of unemployment

Before looking at the data, we review how unemployment is defined in the United States.

The definition of unemployment involves several criteria developed by the Bureau of Labor Statistics (BLS 2023). First, the BLS isolates the part of the population that makes up the working-age civilian noninstitutional population. This group excludes individuals in the armed forces, individuals institutionalized in mental or penal institutions, and children below 16. The resulting group consists of all potential workers residing in one of the 50 US states. Foreigners who reside in the United States are included.

The working-age population is then divided into two categories. The smaller category

is people who are out of the labor force. These people do not have a job and are currently not available or interested to work: people who are retired, studying, or parenting at home.

The larger category is the labor force—everyone who is either working or available and willing to work. People who have a job are counted as employed. Formally, to be employed, you must have worked at least 1 hour in the past week as a paid employee or for your own business. People who are temporarily absent from work, on vacation or because of illness, are also counted as employed.

People who do not have a job are counted as unemployed. From a statistical perspective, people are classified as unemployed when they meet 3 criteria: not currently working; being available to work; and wanting to work. The BLS gauges desire to work by asking respondents if they have been actively searching for a job in the past 4 weeks. Actively searching for a job means using a method that has the potential to result in a job offer, such as contacting an employer, sitting for a job interview, submitting a job application, or using a placement agency. People who are temporarily laid off and expecting to be recalled to their job are also counted as unemployed.

2.2.2. Measurement of unemployment

How is unemployment measured in practice? Unemployment is currently measured through the Current Population Survey (CPS), which is a household survey conducted by BLS. This survey has been running since 1940: it was initially administered by the Census Bureau, and it has been administered by the BLS since 1948. Accordingly, we have a continuous measure of unemployment since January 1948. So, between January 1948 and December 2024, we compute the unemployment rate in the usual way: we take the number of job seekers measured by the BLS (2025n) from CPS, and divide it by the labor force measured by the BLS (2025a) from the CPS. This is the standard, official measure of unemployment rate, labelled U3 by the BLS. Because the unemployment rate measures the share of workers who would like to work but cannot find a job, it is measured as a share of the labor force, not of the entire population.

An issue is that, because the CPS data only start in 1948, we can't see what happened during the Great Depression—the worst recession on record. Fortunately, historians have collected data from different sources that provide a picture of what was happening at that time. Between January 1929 and December 1947, we measure unemployment from the unemployment rate constructed by Petrosky-Nadeau and Zhang (2021). They extrapolate Weir (1992)'s annual unemployment series to a monthly series using various monthly unemployment rates compiled by the NBER.¹

¹There is no monthly measure of unemployment between January 1890 and March 1929. Instead, the monthly unemployment fluctuations are inferred from the spread between the yields of bonds of different quality. Given these limitations, we only begin our analysis in 1929. To be able to start at the beginning of 1929, we use the monthly values produced by Petrosky-Nadeau and Zhang (2021) for January–March 1929, although

The resulting series is graphed in figure 2.1: it depicts the unemployment rate for the United States between January 1929 and December 2024. The shaded areas in the graph correspond to the official recession dates in the United States, which are identified by the Business Cycle Dating Committee of the National Bureau of Economic Research (NBER 2023). The NBER identifies these recessions by looking holistically at numerous macroeconomic variables. Between 1929 and 2024, the NBER identifies 15 recessions.

2.2.3. Prevalence of unemployment

Over the past century, there is always some unemployment in the United States (figure 2.1).

The unemployment rate is countercyclical, rising sharply at the onset of all recessions. During the pandemic recession, the unemployment rate reached its highest level since the end of World War 2: it peaked at 13.0% (2020:Q2). During the Great Recession, the unemployment rate rose to 9.9% (2009:Q4). During the Volcker twin recessions, the unemployment rate rose to 10.7% (1982:Q4). During the first-oil-crisis recession, the unemployment rate reached 8.9% (1975:Q2). The Great Depression was on another scale: the unemployment rate reached 25.3% (1932:Q3). This amount of joblessness is something that we haven't seen since—even during the pandemic recession or Great Recession.

US recessions over the past century have had different sources, but in each one of them the unemployment rate rose. For instance, the recession in 2020 was caused by the coronavirus pandemic. The Great Recession in 2007–2009 coincided with the global financial crisis. The recession in 2001 followed the burst of the dot-com bubble. The recession in 1990–1991 followed the Iraqi invasion of Kuwait and associated oil price shock. The twin recessions in 1980–1982 are associated with the tight monetary policy imposed by Volcker to fight inflation. The recession in 1973–1975 followed the first oil crisis. The biggest recession on record, the Great Depression in 1929–1933, was triggered by the October 1929 stock market crash, followed by a collapse of the banking system. In all these recessions the unemployment rate spiked.

Although there is more unemployment during recessions, unemployment exists at all times. The lowest unemployment rate on record is 1.0%, reached at the end of World War 2 (1944:Q4). This is low but still positive. Such low unemployment was reached because of the war effort: it reduced unemployment through a combination of large-scale military production and large-scale enlistment of potential workers in the armed forces, which shrank the labor force. The World War 2 experience is typical: wartime generally leads to low unemployment. The unemployment rate fell to 2.6% at the end of the Korean War

they are not directly computed from unemployment data. An advantage of using Weir's unemployment series instead of older series is that it counts relief workers, employed by the various New Deal programs, as employed instead of unemployed. This is consistent with the modern definition of unemployment, according to which people with any type of job—relief or not—are counted as employed. By contrast, older series counted these workers as unemployed, adding a handful of percentage points to the unemployment rate (Darby 1976).

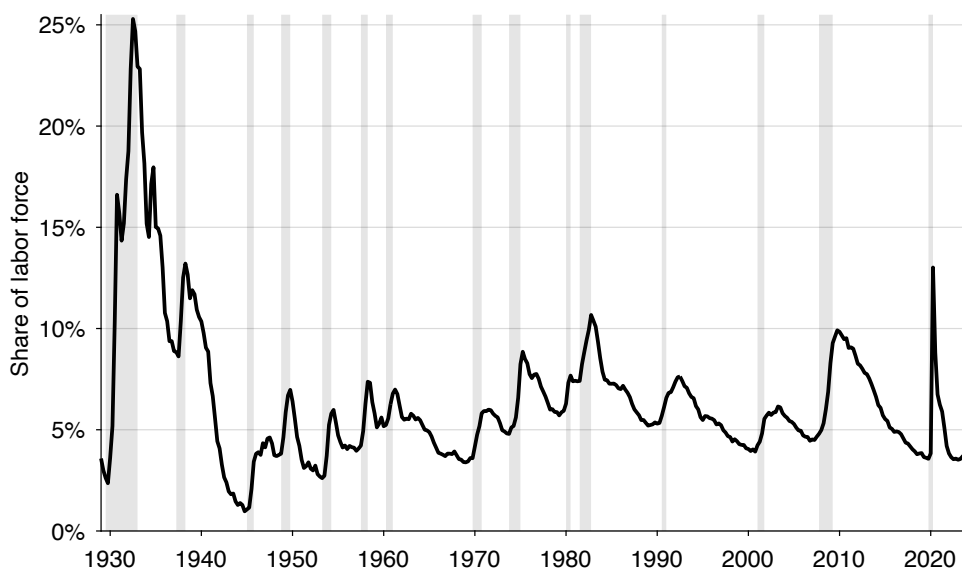


FIGURE 2.1. Unemployment in the United States, 1929–2024

The unemployment rate is a quarterly series computed from monthly, seasonally adjusted data produced by Petrosky-Nadeau and Zhang (2021) and the BLS (2025n,a). Shaded areas indicate recessions dated by the NBER (2023).

(1953:Q2), and it dropped to 3.4% in the middle of the Vietnam War (1969:Q1).

Overall, between 1929 and 2024, there are always some unemployed workers in the US economy. The unemployment rate averages 6.4% over the period. The unemployment rate reached its highest level on record, 25.3%, during the Great Depression and its lowest level on record, 1.0%, at the end of World War 2 (1944:Q4). Over the more stable part of the period, 1948–2019, the unemployment rate averages 5.7%—lower but still very much positive.

2.2.4. Additional measures of unemployment

The concept of unemployment is meant to measure slack on the labor market. So in theory unemployment should comprise anyone who is available and willing to work but unable to find a job. The empirical challenge is to determine who wants to work. Reasonably, the BLS decided that somebody who wants to work is bound to search for a job, so the BLS asks people whether they have been searching for a job to determine whether they are unemployed. However, people search with different intensity and methods, maybe reflecting different willingness to work. The standard unemployment rate (U3) only counts as unemployed people who have been actively searching in the past 4 weeks (searching using a method that has the potential to result in a job offer). So there are workers who

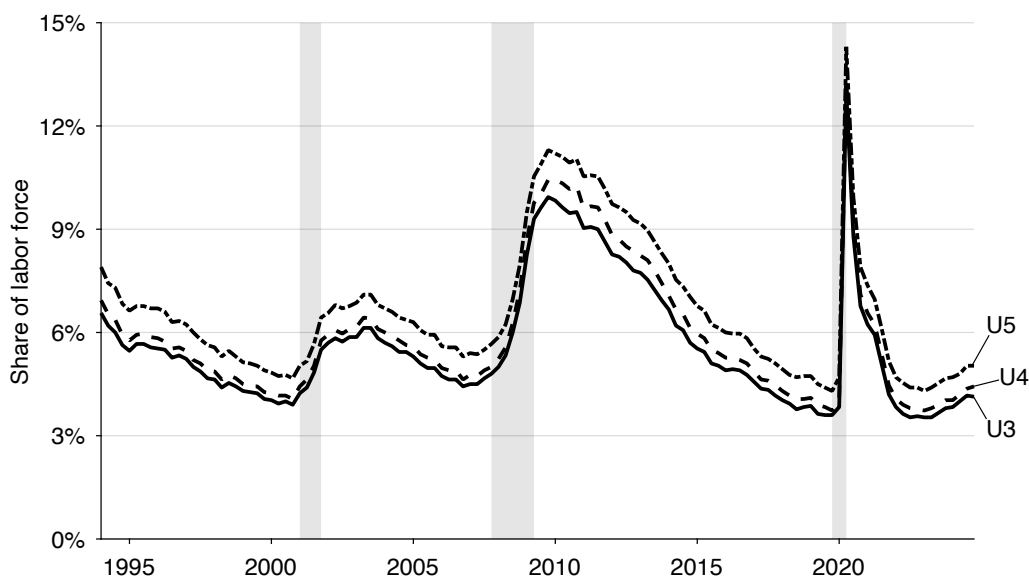


FIGURE 2.2. Alternative measures of unemployment in the United States, 1994–2024

The unemployment rates are quarterly averages of monthly, seasonally adjusted series computed by the BLS (2025o,l,m). Shaded areas indicate recessions dated by the NBER (2023).

have been searching for a job in the past year but not in the past month who are not counted as unemployed. The job seekers might have stopped searching in recent weeks not because they do not want a job but because they could not find any appropriate jobs and lost hope. Similarly, there are people who have been searching for a job passively instead of actively who are not counted as unemployed. These job seekers have been using methods that could not result in a job offer unless additional steps are taken, for instance simply looking at job postings. These people may not have contacted firms because they did not see a good match, or because they were in training. To include all these job seekers into unemployment as well, the BLS developed additional measures of unemployment to count such workers.

The unemployment concept U4 includes everyone in the standard unemployment concept U3, plus discouraged workers. These are people who are available to start a job, have been actively searching for a job in the past 12 months, but have not been actively searching in the past 4 weeks because they became discouraged about their job prospects. When asked why they did not look for work during the last 4 weeks, discouraged workers respond for instance that “There are no jobs available,” or “They have been unable to find work in the past.” They are not counted in U3 because they did not actively search for work in the last 4 weeks.

The unemployment concept U5 includes everyone in U4 plus marginally attached

workers. These are people who are available to start a job, have been actively searching for a job in the past 12 months, but have not been searching in the past 4 weeks for other reasons than discouragement about their job prospects. When asked why they did not look for work during the last 4 weeks, these workers respond for instance that they could not search because of family responsibilities, childcare, or illness. These additional workers are not classified in U3 because they did not actively search for work in the last 4 weeks; they are not classified in U4 because they were not discouraged about their job prospects.

The main takeaway is that the U4 and U5 rates are very close to the standard, U3 rate (figure 2.2). Between 1994 (when the U4 and U5 concepts were introduced) and 2024, the U3 rate averages 5.6%, the U4 rate averages 5.9%, and the U5 rate averages 6.6%. So on average the discouraged workers make up only 0.3% of the labor force, and the marginally attached workers make up just 1% of the labor force. Furthermore, these slivers of the labor force do not vary much over the business cycle. The distance between the U3 and U4 rates is always less than 0.8pp and the distance between the U3 and U5 rates is always less than 1.5pp.

A final form of unemployment is unemployed hours—originating from employees who work part time, although they were available and willing to work full time. Statistically, working part time means working less than 35 hours weekly, while working full time means working 35 hours or more. If a worker wants to work full time but can only find a part-time job, some of the hours that they want to supply are unemployed. However, such workers (part of the U6 concept of unemployment) are only a small sliver of the labor force. In 2024:Q4, only 0.7% of the labor force were working part time because they could not find a full-time job (BLS 2025b,a). And for each of these workers, only a part of their supply of 35 hours was unemployed.

2.3. Prevalence of slack on the product market

We have seen that slack is prevalent on the labor market. But slack is not limited to the labor market; it is also prevalent on the product market. Indeed, producers are never able to sell all the goods and services that they could potentially produce to customers.

2.3.1. Prevalence of unsold goods

Slack takes several forms on the product market. The most basic form is unsold goods, such as unsold food. Unsold food by farmers and manufacturers, and in grocery stores and restaurants, generate a vast amount of food waste in the United States (ReFED 2025). In 2023, 31% of the US food capacity (in tons) is surplus: either unsold or unused by businesses, or uneaten at home. The value of the food unsold by farms, manufacturers, restaurants, and grocery stores represents 5% of the value sold.

It is not only food that US firms have difficulties selling; any good is difficult to sell. When firms produce durable goods that they cannot sell, they generate unsold inventory. Indeed if firms keep producing durable goods when there is less demand for them, the unsold goods pile up in inventory. We do see inventories swell up when firms are unable to sell their goods, and inventories shrink when demand is brisk (Bils and Kahn 2000).

2.3.2. Prevalence of idle capacity

Another form of slack on the product market is idle capacity in firms: capacity that is available and ready to produce goods and services, but remains unused because of a lack of demand. It is a type of slack that is less commonly discussed but still very prevalent, as we see from data collected by the Institute for Supply Management (ISM 2025). The ISM keeps track of the “operating rate”, both for manufacturing and nonmanufacturing industries. The operating rate indicates the actual production level of firms as a fraction of the maximum production level that these firms could have achieved given their current capital and labor. From this operating rate, we can create a rate of idleness, which is just $1 - \text{operating rate}$. The idleness rate is the share of productive capacity that is idle because of a lack of demand: the amount of goods or services not sellable and therefore not produced, as a fraction of the total capacity of a firm given current capital and labor.

In the United States, we find that some capacity is idle at all times (figure 2.3). Between 1990 and 2024, the rate of idleness in the manufacturing sector averages 17.3%. And between 2000 and 2024, the rate of idleness in nonmanufacturing sectors averages 13.9%. This means that usually, given their capital and labor, firms could sell about 15% more than what they actually sell, if they had the demand for it.

The rate of idleness is the product-market equivalent of the rate of unemployment. Perhaps surprisingly, the rates of idleness in the manufacturing and nonmanufacturing sectors are much higher than the rate of unemployment. There are several ways to see this. The average rates of idleness are higher than the average rate of unemployment between 1990 and 2024, which is only 5.7%. The manufacturing rate of idleness is always above 10% and the nonmanufacturing rate of idleness is always above 8%. Both lower bounds on idleness correspond to already significant amounts of unemployment. Last, we can compare figure 2.1 to figure 2.3: we can see that at any time the unemployment rate is below the rates of idleness. The data show us that more labor is idle within firms than outside them: product market slack is a critical phenomenon.

Just like the rate of unemployment, the rates of idleness are sharply countercyclical, climbing in recessions. Idleness in the manufacturing sector peaked at 33.0%, during the Great Recession (2009:Q2). Manufacturing idleness was also elevated during the other recessions in the data: the idleness rate reached 20.6% after the Iraq War recession (1991:Q2), 22.5% after the dot-com bubble recession (2001:Q4), and 24.1% after the pandemic re-

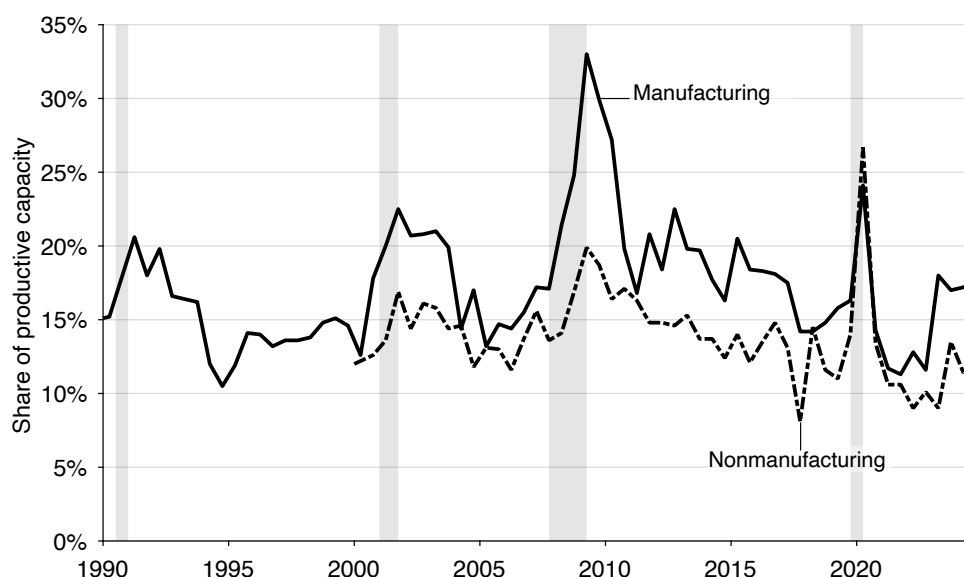


FIGURE 2.3. Idle capacity in the United States, 1990–2024

The idleness rates are quarterly series constructed by interpolating the semiannual operating rates measured by the ISM (2025). For nonmanufacturing sectors, the operating rate is only available after 2000. Shaded areas indicate recessions dated by the NBER (2023).

cession (2020:Q2). Slack in the nonmanufacturing sectors peaked at 26.7%, during the pandemic recession (2020:Q2). Nonmanufacturing idleness also reached 19.9% after the Great Recession (2009:Q2) and 16.9% after the dot-com bubble recession (2001:Q4). We see that idleness is a much bigger problem during recessions. We also see that the Great Recession was relatively worse for manufacturing sectors, while the pandemic recession was relatively worse for nonmanufacturing sectors.

2.3.3. When product market slack bleeds into the labor market

We have just seen that people employed by firms are idle some of the time at work because firms do not have enough business. In that case product market slack materializes as idleness at work. If workers are sent home by firms and their hours reduced, then product market slack materializes in another way: as full-time workers being forced to work part time because their employers do not have sufficiently many customers. These workers are counted as employed, but they are flagged as involuntary part-time workers in the CPS by the (BLS 2025c). Their usual hours of work are above 35 hours a week but they are now forced to work less than 35 hours because of unfavorable business conditions or low seasonal demand.

Again, this is a form of product market slack, since the issue is that the workers' labor

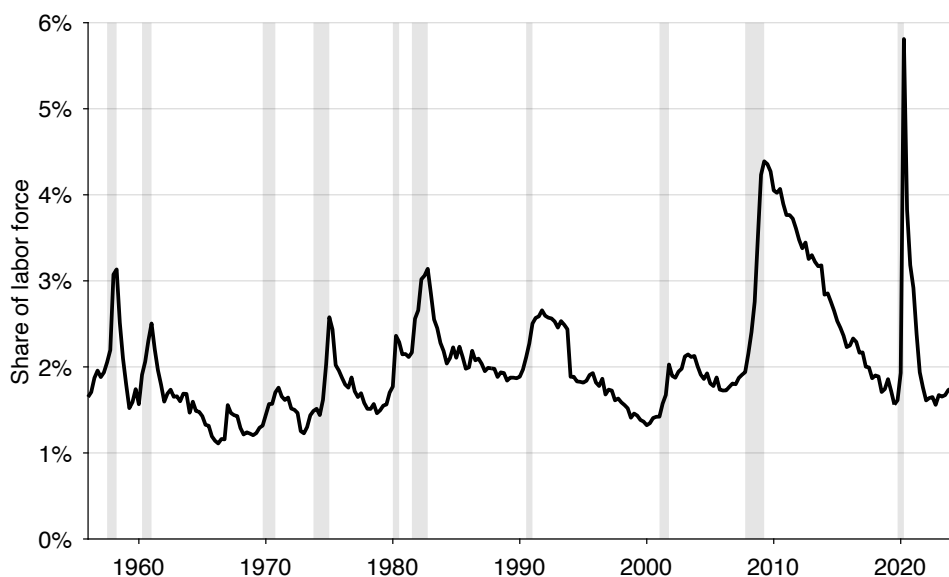


FIGURE 2.4. Workers forced to reduce hours because of product market slack in the United States, 1956–2024

The share of workers forced to reduce hours is a quarterly series computed from monthly, seasonally adjusted data produced by the BLS (2025c,a). Shaded areas indicate recessions dated by the NBER (2023).

services cannot be sold to customers—not that workers cannot sell their labor to firms. These workers are another component of the U6 unemployment concept measured by the (BLS 2023), but they are best interpreted as product market slack.

We see that many people are working part-time because of slack on the product market (figure 2.4). On average between 1956 and 2024, 2.1% of the labor force is working less than full time because of product market slack; that share is always above 1%. Just like the slack rate in figure 2.3, that share of workers is sharply countercyclical: much higher in bad times than in good times. During the Great Recession, the share spiked to 4.4% (2009:Q2); during the pandemic recession, it spiked to 5.8% (2020:Q2).

2.4. Coexistence of sellers and buyers

We have observed slack on the labor market in the form of unemployed workers and slack on the product market in the form of idle capacity. To explain these patterns, we could simply use a nonclearing Walrasian model—a Walrasian model with a non-market-clearing price.² With an above-market-clearing price in the product market, there would be excess supply of goods and services, and we could explain why there are firms that

²Benassy (1993) offers a wonderful survey of that literature.

have goods and services that are not sold. With an above-market-clearing wage in the labor market, there would be excess supply of labor, and we could explain why there are workers who want to work but are not employed.

However, the world is more complicated than that because just as there are sellers who are unable to sell their goods or services, we see buyers who are unable to purchase goods or services. The coexistence of sellers and buyers is inconsistent with the nonclearing Walrasian model. In a situation of excess supply, such a model can explain the existence of slack. But in excess supply, buyers are on their demand curves, so they are able to purchase the amount of goods or services that they desire at the prevailing price. In other words, all buyers can successfully and seamlessly purchase the amount of goods and services that they desire.

We now review evidence of this conundrum. We start with the labor market, where the presence of job vacancies has been well documented. We then turn to the product market, where the evidence is less readily available. In the book, to explain the coexistence of sellers and buyers that are unable to trade, we will organize each market around a matching function. The matching function will mediate all trades on the market, capturing the difficulties that sellers and buyers alike face in finding each other.

2.4.1. Coexistence of job seekers and vacant jobs

On the labor market, we always have unemployed workers trying to find jobs, as we have seen, but we also always have firms trying to fill vacant jobs. The presence of vacant jobs is the concrete manifestation that firms are looking to hire labor.

How can we assess whether firms want to hire workers at any point in time? Well, we have to look at the number of job vacancies that firms report. Job vacancies are measured through the Job Openings and Labor Turnover Survey (JOLTS), which is a firm survey conducted by the BLS since January 2001.³ The definition that the BLS (2024) adopted for job vacancy parallels the definition of an unemployed worker. There are three conditions. First, a specific position exists, and there is work available for that position. Second, the job must start within one month. And third, there is active recruiting for that position for workers outside the establishment. So each vacant position is associated with recruiting within firms, which ensures that firms are actively looking for new workers. Hence, between January 2001 and December 2024, we compute the vacancy rate from the number of job openings measured by the BLS (2025d) from the JOLTS. We divide the number of

³To best align CPS and JOLTS data, we shift forward by one month the number of job openings from JOLTS. For instance, we would assign to October 2024 the number of job openings that the BLS assigns to September 2024. The motivation for this shift is that the number of job openings from the JOLTS refers to the last business day of the month (Monday 30 September 2024), while the labor force from the CPS refers to the Sunday–Saturday week including the 12th of the month (Sunday 6 October 2024 to Saturday 12 October 2024). So the number of job openings refers to a day that is closer to next month’s CPS week than to the current month’s CPS week. With this shift, the JOLTS conveniently starts in January 2001 instead of December 2000.

job openings by the size of the labor force, which is measured by the BLS (2025a) from the CPS, to express the number of vacant jobs as a share of the labor force.

There are no governmental measures of job vacancies in the United States before 2001, but other organizations have been collecting data on vacant jobs before that, and researchers have transformed these data into vacancy series. Between January 1951 and December 2000, we use the vacancy rate produced by Barnichon (2010) from the Conference Board's help-wanted advertising index. The Conference Board index aggregates help-wanted advertising in major metropolitan US newspapers. Given the shift from print advertising to online advertising with the advent of the internet, the Conference Board index needs to be adjusted in the 1990s, which Barnichon (2010) has done. Between January 1929 and December 1950, we use the vacancy rate produced by Petrosky-Nadeau and Zhang (2021) from MetLife's help-wanted index. This index aggregates help-wanted advertisements from newspapers across major US cities, and reliably proxies for job vacancies (Zagorsky 1998).⁴

A first natural question is whether help-wanted indices proxy well for job vacancies. Comparing them to local data on job vacancies and other available evidence, (Abraham 1987) and (Zagorsky 1998) find that both the Conference Board index and the MetLife index proxy well for job vacancies. A second natural question is how the number of job vacancies can be recovered from an index? This is done by scaling the indexes to eventually match the number of job vacancies at the beginning of the JOLTS: the Conference Board index is scaled to align with the JOLTS vacancy rate in January 2001, and the MetLife index is scaled to align with the scaled Conference Board index in January 1951. Scaling effectively translates the indexes into vacancy rates.

Next, we splice the three vacancy series to create a continuous vacancy rate covering January 1929–December 2024 (figure 2.5). The series is procyclical, rising sharply in expansions. In the recovery from the coronavirus pandemic, the vacancy rate reached its highest level on record: it peaked at 7.2% (2022:Q2). Just before the Great Recession, the vacancy rate reached 3.1% (2007:Q2). Just before the burst of the dot-com bubble, the vacancy rate peaked at 4.1% (2000:Q1). Before the Volcker recessions, the vacancy rate was even higher, spiking at 5.1% (1979:Q2). Job vacancies are generally elevated in wartime: the vacancy rate reached 6.7% at the end of World War 2 (1945:Q1), 4.3% during the Korean War (1953:Q1), and 5.1% during the Vietnam War (1969:Q1).

While it is true that job vacancies exist in large numbers during expansions, vacant jobs exist at all times, which tells us that firms are always looking for some workers, just like some workers are always looking for jobs. Even in recessions, the vacancy rate remains positive. The lowest vacancy rate on record is 0.7%, reached in the depth of the Great Depression (1933:Q1). This is a very low number, but still positive. During the Great

⁴Petrosky-Nadeau and Zhang (2021) produce a vacancy series that starts in 1919, but we start the vacancy series in 1929 to align it with the unemployment data from figure 2.1.

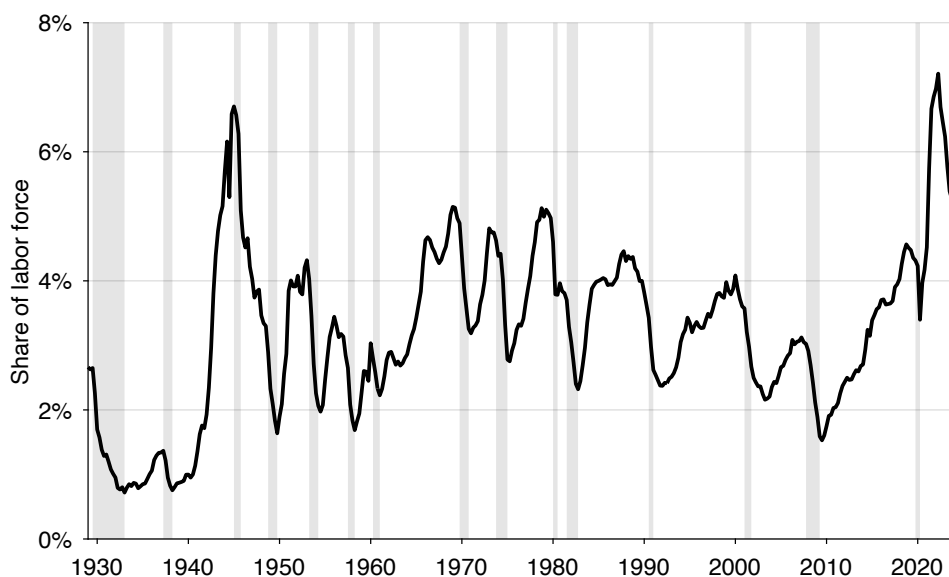


FIGURE 2.5. Job vacancies in the United States, 1929–2024

The vacancy rate is a quarterly series computed from monthly, seasonally adjusted data produced by Petrosky-Nadeau and Zhang (2021), Barnichon (2010), and the BLS (2025d,a). Shaded areas indicate recessions dated by the NBER (2023).

Recession, the vacancy rate dwindled to 1.5% (2009:Q3). The vacancy rate bottomed at 2.2% after the dot-com bubble recession (2003:Q2) and at 2.3% after the Volcker twin recessions (1982:Q4). The vacancy rate was also extremely low in the recessions that followed the Great Depression: 0.8% after the 1937–1938 recession (1938:Q2) and 1.6% after the 1948–1949 recession (1949:Q4).

Overall, between 1929 and 2024, the vacancy rate averages 3.2%. The vacancy rate reached its highest level on record, 7.2%, during the recovery from the pandemic and its lowest level on record, 0.7%, during the Great Depression. The bottom line is that, just as there always are unemployed workers, there always are vacant jobs.

2.4.2. Beveridge curve

We have seen in figures 2.1 and 2.5 that the number of job seekers is countercyclical while the number of vacant jobs is procyclical. This implies that the unemployment and vacancy rates are negatively related. This negative relationship between unemployment and vacancy rates corresponds to the Beveridge (1944) curve: a key empirical relationship that guides our modeling and policy design throughout the book.

The Beveridge curve appears clearly on a scatter plot with unemployment rate on the horizontal axis and vacancy rate on the vertical axis (figure 2.6). Here to illustrate, we

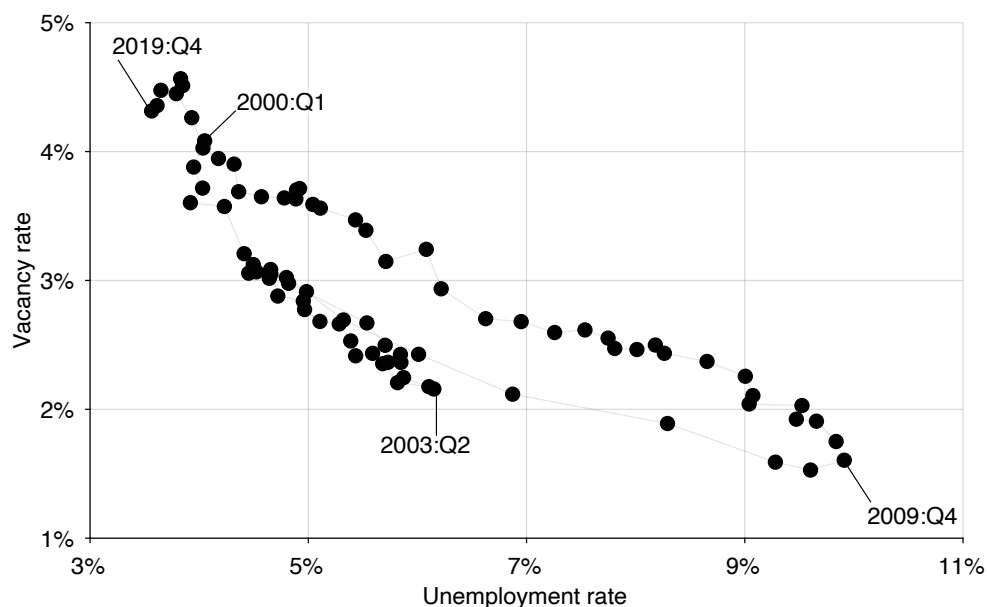


FIGURE 2.6. Beveridge curve in the United States in recent decades, 2000–2019

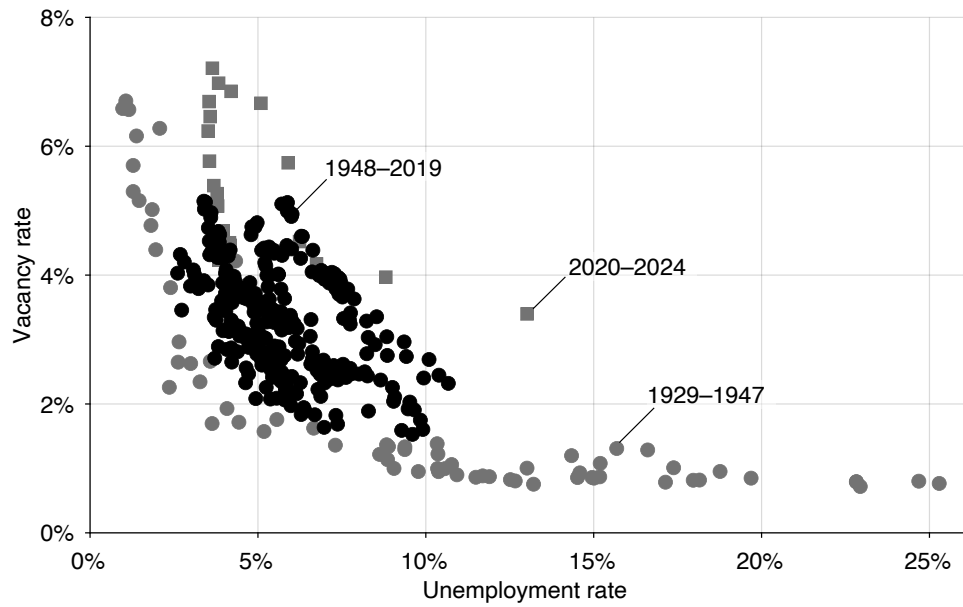
The unemployment rate comes from figure 2.1 while the vacancy rate comes from figure 2.5.

only use recent data, between 2000 and 2019. We stop before the pandemic to simplify the picture. The Beveridge curve says that when the economy is in a slump, there are a lot of job seekers and few vacant jobs. Good examples of such situations are 2003:Q2, when the unemployment rate reached 6.2% and the vacancy rate fell to 2.2%, and 2009:Q4, when the unemployment rate peaked at 9.9% and the vacancy rate plummeted to 1.6%. Conversely, when the economy is in a boom, there are few job seekers and many vacant jobs. Such a situation occurred in 2019:Q4, when the unemployment rate dipped to 3.8% and the vacancy rate reached 4.3%.

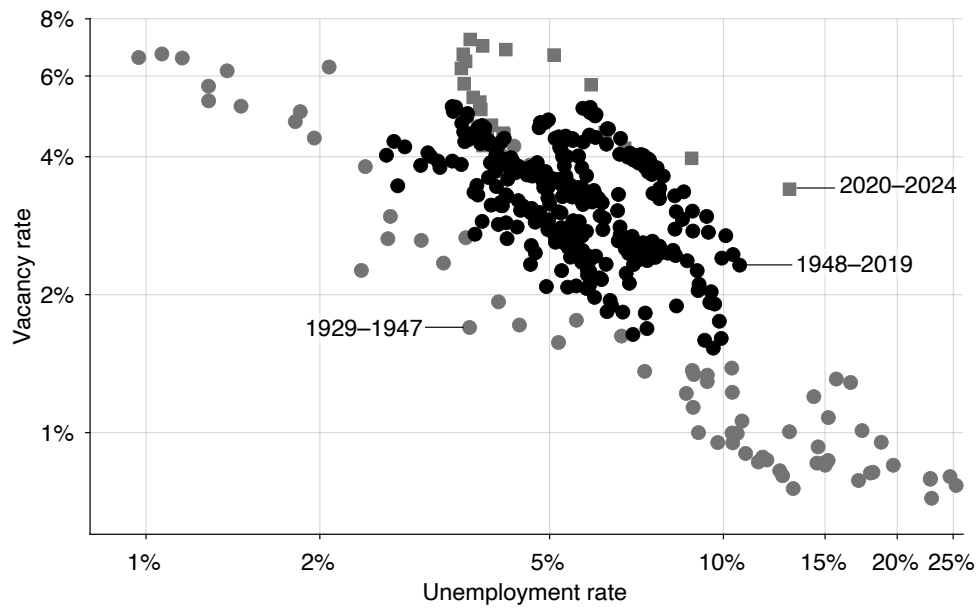
Although the Beveridge curve was first observed in Great Britain (Dow and Dicks-Mireaux 1958), it holds remarkably well in the United States (Blanchard and Diamond 1989; Elsby, Michaels, and Ratner 2015).

To illustrate, figure 2.7 displays the Beveridge curve for the entire period of our data, 1929–2024. The figure offers two displays: a regular display (figure 2.7A) and a display on a log scale (figure 2.7B), which is useful to show that the Beveridge curve is isoelastic—that there is a linear relationship between log vacancy and log unemployment.

The figure distinguishes three subperiods. The initial period, 1929–1947, was extremely volatile, not least because it experienced two dramatic events, the Great Depression and World War 2. Data available for the period are also the noisiest since neither unemployment nor job vacancies were measured by the government. Around the middle of the Great Depression and at the end of World War 2, the Beveridge curve is subject to two



A. Regular scale



B. Logarithmic scale

FIGURE 2.7. Beveridge curve in the United States, 1929–2024

The unemployment rate comes from figure 2.1 while the vacancy rate comes from figure 2.5.

large, outward shifts. These shifts are hard to discern on this figure but will be seen clearly in chapter 13.

In the next, postwar period, 1948–2019, the Beveridge curve is well-behaved. The US Beveridge curve is stable for long periods of time, before shifting inward or outward after a decade or more (Michaillat and Saez 2021, figure 5). The structural breaks in the curve are of modest size compared to those that occurred in the 1929–1947 period. The economy is always tightly on the Beveridge curve.

Finally, the last subperiod corresponds to the coronavirus pandemic and its aftermath, 2020–2024. The pandemic shattered the US labor market and dramatically shifted the Beveridge curve outward. The Beveridge curve slowly recovered from the pandemic shift over the next few years and it only came back close to its pre-pandemic position in 2024.

In fact, unemployment and vacancy rates appear not only to be negatively related, but to be the inverse of each other. So doubling the unemployment rate cuts the vacancy rate in half, and conversely, doubling the vacancy rate cuts the unemployment rate in half. Mathematically, the property that the unemployment rate and vacancy rate are inversely related implies that the Beveridge curve is a rectangular hyperbola:

$$vu = A,$$

where $A \in (0, 1)$ is a constant. The rectangular hyperbola can be rewritten $v = A/u$, so we can see that the elasticity of the vacancy rate with respect to the unemployment rate along the Beveridge curve is -1 .

It is possible to establish that the Beveridge curve is close to a rectangular hyperbola formally. This can be done by estimating the elasticity of the vacancy rate with respect to the unemployment rate along each of its branch. An elasticity of -1 corresponds to a rectangular hyperbola. It turns out that during the postwar period, the elasticity of the Beveridge curve on each branch remains between -0.84 and -1.02 , so never far from -1 (Michaillat and Saez 2021, figure 6).

It is possible to make the case for a hyperbolic Beveridge curve without introducing structural breaks and looking to branches in isolation. The idea is to remove a very slow-moving trend from the unemployment and vacancy rates to eliminate the decennial movements of the Beveridge curve, and to then estimate the elasticity of the detrended vacancy rate with respect to the detrended unemployment rate. To be precise, we remove from log unemployment rate a slow-moving trend obtained from a Hodrick-Prescott (HP) filter with smoothing parameter 10,000.⁵ We do the same to the log vacancy rate. We then regress detrended log vacancy rate on detrended log unemployment rate. We focus on the 1948–2019 period because the large shifts in the Beveridge curve before and after

⁵Appendix A provides an introduction to the HP filter and explains the choice of the smoothing parameter. We use the HP filter with that smoothing parameter throughout the book, whenever detrending is required.

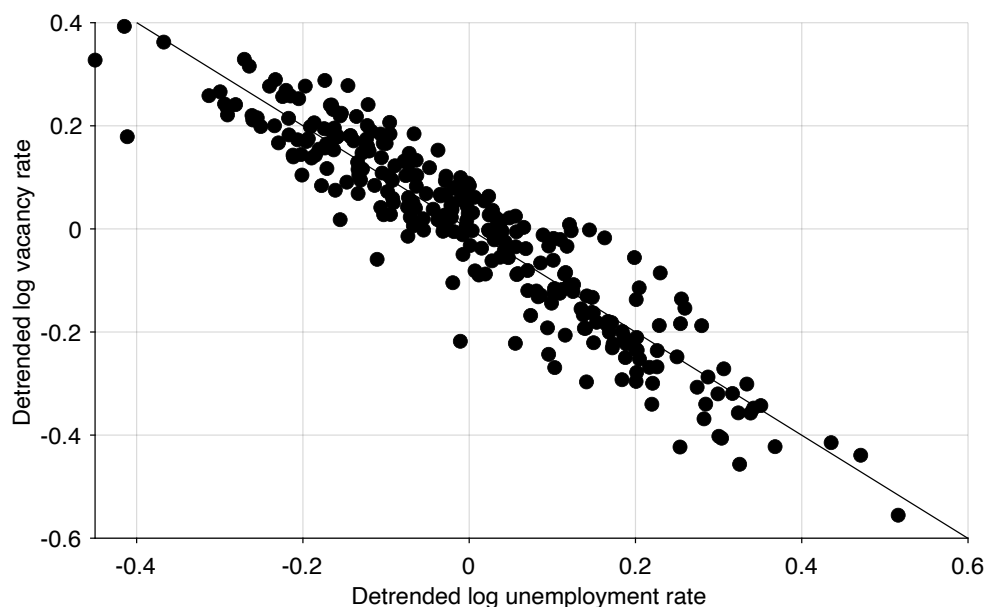


FIGURE 2.8. The US Beveridge curve is a rectangular hyperbola

The unemployment rate comes from figure 2.1 while the vacancy rate comes from figure 2.5. The figure plots the logarithm of the vacancy rate against the logarithm of the unemployment rate, both detrended using an HP filter with smoothing parameter 10,000. The data are for 1948–2019.

that period are difficult to handle. We find that the slope is exactly -1.00 with a good fit, $R^2 = 0.87$ (figure 2.8). Thus, with detrended variables, the US Beveridge curve is a perfect rectangular hyperbola over seven decades, 1948–2019.

2.4.3. Coexistence of product sellers and buyers

On the product market, we have seen that sellers are continuously trying to sell goods and services; at the same time, buyers are continuously trying to purchase them.

Every day, US consumers spend a significant amount of time trying to buy goods and services: researching their shopping, shopping online and in person, driving to shops, and waiting in line at shops.

A first reason why buying takes time is that it is difficult to find the appropriate good that fits one's needs. This difficulty manifests itself in the large number of goods that are returned after purchase because they did not meet the buyer's needs. In 2020 in the United States, \$428 billion of merchandise were returned to sellers—a value representing 10% of retail sales (Janssen and Williams 2024, p. 387). For online retail only, \$102 billion of merchandise were returned, representing 18% of sales.

Another reason why shopping takes time is that goods are not always available in shops. Using the microdata underlying the Consumer Price Index, Bills (2016, table 1) finds

that temporary stockouts are quite common: the average stockout rate is 4.6% across 180 categories of goods between 1988 and 2009. That is, about 5% of visits to a store do not result in a purchase because the desired product is unavailable. Taylor and Fawcett (2001, figure 1) surveyed the availability of 40 items (half advertised and half not advertised) across 20 US stores for 4 weeks. They find that for advertised items, stockouts occur on 12%–16.5% of the visits; for regular items, stockouts occur on 6.1%–7.6% of the visits. Stockouts are also prevalent for online retailers. Jing and Lewis (2011, table 1) find that 25.4% of orders placed to an online retailer were imperfectly filled—some of the items that were ordered could not be shipped to the customer. The value of the items in stockout represents 9.7% of the total value of the orders.

The activity of purchasing goods and services is not limited to households. Firms are also constantly trying to purchase goods and services from other firms. In fact, the amount of labor devoted by firms to source goods and services might not be very different from the amount of labor devoted to recruiting workers. According to the Occupational Employment Survey (OES) run by the BLS, on average between 1997 and 2012, 560,600 workers were employed in buying, purchasing, and procurement occupations while 543,200 workers were employed in occupations involving recruitment, placement, screening, and interviewing (Michaillat and Saez 2015, figure 2B).

2.5. Prevalence of long-term relationships

The difficulties in matching sellers and buyers that create slack also incentivize sellers and buyers to form long-term relationships once they have found each other. Indeed, once buyer and seller are locked into a long-term relationship, they can trade seamlessly, without having to find a new trading partners each time. In this section we show that long-term relationships are prevalent on the labor and product markets, providing indirect evidence of underlying matching difficulties.

2.5.1. Long-term relationships on the labor market

US firms typically employ workers on long-term contracts. Kalleberg (2001, table 8.2) finds that in 1997, 71% of men and 62% of women are engaged in standard employment, in which the work is full time and the relationship is assumed to continue for a substantial period or indefinitely. In addition, 8% of men and 22% of women are engaged in regular part-time employment, which is also expected to continue for a substantial period or indefinitely. So overall 79% of employed men and 84% of employed women are in long-term employment relationships. Workers who are not in long-term employment are for instance independent contractors (although these workers might be in long-term relationships with their customers) or temporary help agency workers. These numbers

have not changed much between the 1960s and 1990s (Kalleberg 2001, table 8.3). Between the 1990s and 2010s, the number of workers in alternative work arrangements increased by a few percentage points, driven partly by the emergence of the gig economy (Katz and Krueger 2019, table 2).

To assess how long employment relationships generally last, we turn to the job-separation rate. The rate indicates the number of worker-firm relationships that end in a given month, as a share of the total number of existing relationships. It is computed by the (BLS 2025k) from JOLTS data since 2001. As earlier, we omit data from the coronavirus pandemic and its aftermath because they are not representative of the typical behavior of the US labor market (the pandemic led to a dramatic, unprecedented increase in job separations).

Overall, we see that the job-separation rate is not that large, which tells us that people keep their jobs for quite a long time (figure 2.9A). Between 2001 and 2019, the monthly job-separation rate averages 3.6%. This means that every month, 3.6% of workers are willingly or unwillingly leaving their job. Assuming that job separations are a Poisson process, we infer that the average duration of a job is $1/0.036 = 27.8$ months, or about 2.3 years.

A Poisson process is memoryless, which means that irrespective of how long a worker has been with the firm, conditional on being employed today, they can expect to be there for about 2 more years. In the US labor market, however, memorylessness does not hold. Initially, when a worker starts a new job, the separation risk is quite high. Then, as a worker's tenure with the firm increases, the separation risk falls drastically. As a result, workers who have stayed a while with a firm are likely to stay much longer still—they are likely to spend their career at the firm.

Indeed, Hall (1982) found that in the US in the 1960s and 1970s, the average worker kept their job for about 8 years, and 25% of workers held jobs that would last at least 20 years. Hall (1982, p. 724) explains that “The jobs held by middle-aged men are remarkably stable.... Monthly separation rates, which are of the order of 3% for workers in general, are about 0.25% for middle-aged men with at least ten years on the job.” Moreover, the duration of US jobs has remained remarkably stable since the 1980s (Molloy, Smith, and Wozniak 2024, figures 1 and 2).

An interesting aspect of the job-separation rate is that it is stable over the business cycle. People typically assume that waves of job separations occur at the onset of each recession, but in recent decades, that has not really been the case: if anything the job-separation rate dropped during the course of the dot-com recession and Great Recession. How can that be? Don't firms lay workers off at the onset of recessions? It used to be true that the layoff rate would spike at the onset of recessions (Akerlof, Rose, and Yellen 1988, figure 2). But between 2001 and 2019, the layoff rate increases only slightly during

recessions (figure 2.9B). Moreover, the rise in layoffs in recessions is offset by a drop in quits (see figure 2.9B and Akerlof, Rose, and Yellen 1988, figure 1). Overall, because quits (initiated by workers) are procyclical while layoffs (initiated by firms) are countercyclical, total job separations are close to acyclical.

2.5.2. Long-term relationships on the product market

Although it is maybe less well known, long-term relationships are also extremely common on the product market. Blinder et al. (1998, Table 4.12) report that on average, 85.2% of sales by US firms go to long-term customers—either other businesses or consumers.

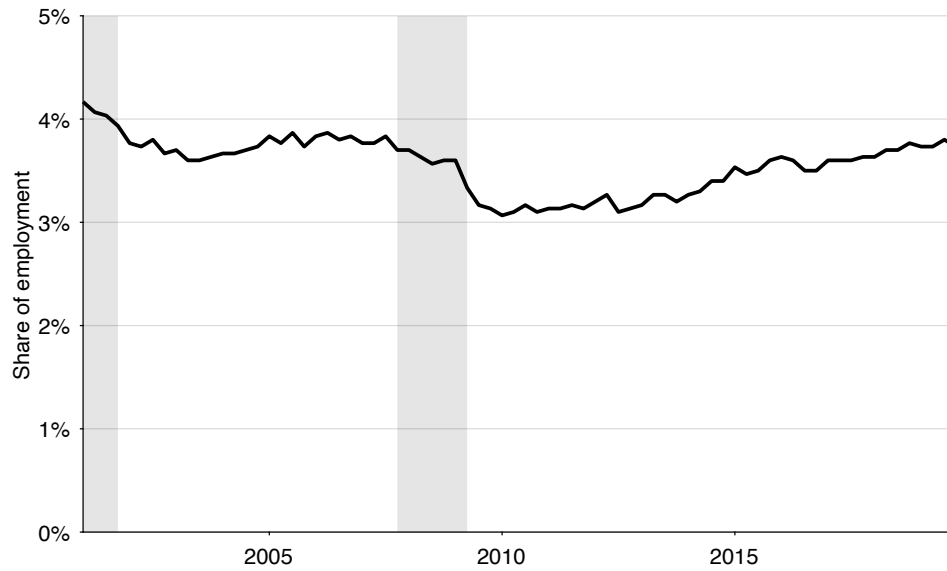
Indeed, long-term relationships are exceedingly common in business-to-business transactions. A broad share of business-to-business transactions are conducted under explicit contracts that link sellers and buyers over time. In BLS data on contractual arrangements between firms trading intermediate goods, Goldberg and Hellerstein (2011) find that one-third of all transactions are conducted under contract across industries, for both goods and services. In data on foreign trade and transactions from the Census Bureau, Monarch and Schmidt-Eisenlohr (2023, table 1) find that more than 80% of US imports occur through long-term relationships. Fernandez-Villaverde et al. (2025, pp. 2509–2510) confirm that most relationships between firms and their suppliers in the United States are long term: the average customer-supplier relationship has been ongoing for 3.5 years.

Long-term relationships between consumers and sellers are also common. Dube, Hitsch, and Rossi (2010) documents that consumers have a higher probability of choosing products that they have purchased in the past—thus remaining loyal to the product and brand that they patronized in the past. But, as Okun (1981, pp. 150–151) observes, personal relationships with sellers are even more important than relationships with brands:

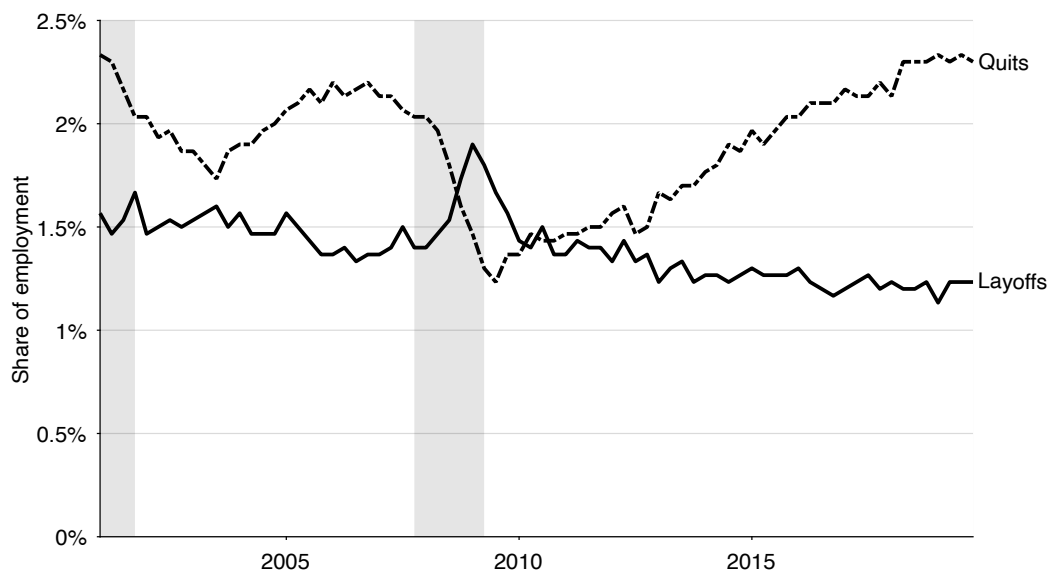
The model of customer-supplier attachments applies most obviously to suppliers of services that are not easily evaluated by shopping. The resulting list is long and important in terms of the consumer's budget, including medical practitioners like the physician, dentist, and veterinarian; personal services like those of the barber, beautician, and dry cleaner; repair services of the automobile mechanic, house painter, plumber, and electrician; financial service suppliers like the lawyer, accountant, insurance agent, and branch banker. The considerations extend to goods as well as services ... the consumer wants a reliable optician, pharmacist, antique dealer, butcher, and clothier.

2.6. Cyclical slack

This book is about business cycles. In it we will see that slack not only exists but also fluctuates. And fluctuations in slack are a central part of business cycles.



A. All job separations



B. Quits and layoffs

FIGURE 2.9. Monthly job separations in the United States, 2001–2019

The job-separation rates are quarterly averages of monthly, seasonally adjusted series computed by the BLS (2025k,h,j). Shaded areas indicate recessions dated by the NBER (2023).

But what are business cycles? Here is a standard definition of business cycles, proposed by Burns and Mitchell (1946, p. 3):

A cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle; this sequence of changes is recurrent but not periodic; in duration business cycles vary from more than one year to ten or twelve years; they are not divisible into shorter cycles of similar character with amplitudes approximating their own.

The typical measures of activity used to describe business cycles include industrial production, output, and consumption. They all move in tandem, deviating from trend by a few percent, up or down, at each expansion and recession (Stock and Watson 1999).

The business cycle fluctuations to be explained are presented on figure 2.10. We recognize the typical pattern: output and consumption are above trend in expansion, and below trend in recessions. In the 21st century, output and consumption move essentially in tandem (this was not necessarily the case before). Between 2000 and 2024, the correlation between detrended output and consumption is 0.92.

Figure 2.10 addresses a possible concern about the approach taken in the book. Output is the sum of private and public consumption, private and public investment, and net exports. The models in the book abstract from imports and exports, and from capital and investment. While one might worry that a model abstracting from trade and investment could not explain output fluctuations, the figure shows that private consumption moves so closely with output that our focus is well-suited to the task.

The goal of business cycle research is to develop a model that can explain the fluctuations described by Stock and Watson (1999). The pervasiveness and cyclical nature of slack suggest that business cycles may reflect fluctuations in slack in addition to fluctuations in productive capacity. Indeed, by definition, slack is the share of the economy's productive capacity that is either not used or not sold, so

$$(2.1) \quad \text{output} = (1 - \text{slack}) \times \text{productive capacity}.$$

This simple slack identity shows that there are two possible and nonexclusive sources of output fluctuations: fluctuations in productive capacity and fluctuations in slack.

There are three inputs into productive capacity. First is the capital stock. Second is the labor force: the share of the working-age civilian population that is available and willing to work. Third is technology: the blueprint to the production process, which shows how to combine capital and labor effectively. These inputs are combined through a production process to generate the productive capacity. We will see in the next sections that all these inputs, and therefore productive capacity, are broadly acyclical. This will leave fluctuations

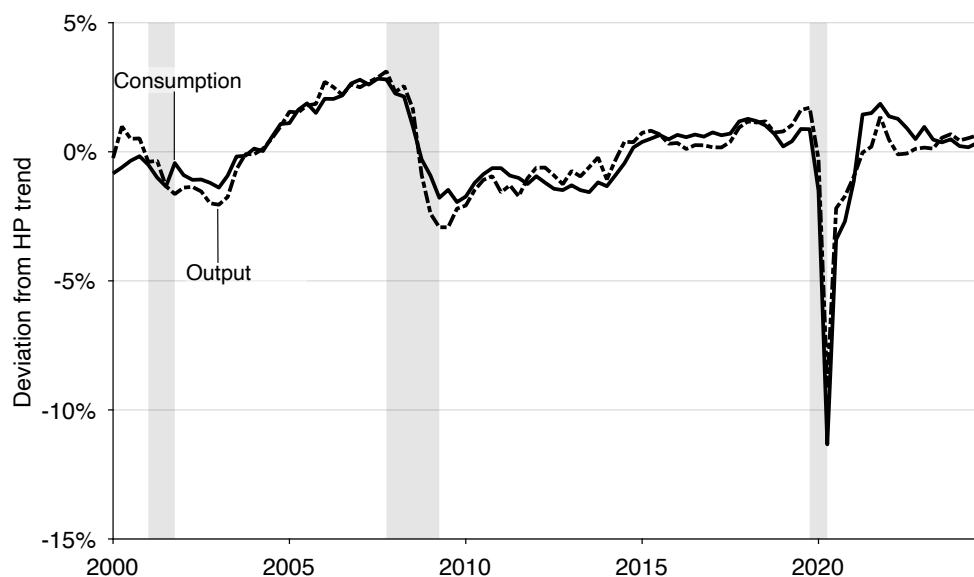


FIGURE 2.10. Fluctuations in consumption and output in the United States, 2000–2024

Output is real gross domestic product measured by the BEA (2025b). Consumption is real personal consumption expenditure measured by the BEA (2025c). The original quarterly, seasonally adjusted series are detrended by applying to their logarithms a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).

in slack as the main explanation for business cycle fluctuations.

2.6.1. Capital is broadly acyclical

How much does the capital stock vary over time and, in particular, over the business cycle? This is a fairly easy question to answer: we just need to look at capital stock at constant national prices for the United States graphed from 1950 to 2019 (University of Groningen and University of California–Davis 2023). It is clear that the capital stock is growing over time, but notice that the growth of the capital stock is extremely smooth—it is essentially unaffected by recessions or by the business cycle in general. This is not a new finding. If we go back to the first chapter of the main textbook of the Real Business Cycle literature, *Frontiers of Business Cycle Research*, one of the findings is that capital stock fluctuates much less than output and is largely uncorrelated with output (Cooley and Prescott 1995). Thus, we conclude that the US capital stock is acyclical.

2.6.2. Labor force is broadly acyclical

Next, how much does labor supply vary over the business cycle? The labor force is the labor supply. It is composed of the members of the working-age civilian population who

want to work. Mathematically, the labor force is the product of the working-age civilian population and the labor force participation rate.

The working-age civilian population, of course, is growing at a very stable rate and not subject to cyclical fluctuations (BLS 2025i). What we need to look at is the share of the civilian population that wants to work: the labor force participation rate. This is the main decision that people make in the real world: whether to participate in the labor force or not.

It turns out that the participation rate doesn't really vary over the business cycle (BLS 2025e). The participation rate is molded by medium-run fluctuations that do not have much to do with the business cycle. Until the 1970s, the rate was fairly stable with some small fluctuations. There was then a big increase in the participation rate between the mid 1960s and the mid 1990s as women entered the labor force in large numbers (BLS 2025g). After the mid 1990s, there was a plateau followed by a decline in the participation rate from the 2000s. The participation rate for men has been declining since the 1950s, but this decline was more than compensated until the 1990s by women entering the labor force (BLS 2025f).

The medium-run forces might make it difficult to ascertain that the labor force participation rate is truly acyclical—although by looking separately at the participation rates for men and women, it is quite clear that they are acyclical (BLS 2025f,g). In any case, to be completely sure, researchers have explored the cyclicity of the participation rate with statistical techniques. Using US data covering 1946–1954, Rees (1957, p. 32) does not find evidence of the discouraged-jobseeker theory. In US data covering 1960–2006, Shimer (2009, p. 294) finds that the labor force participation rate is acyclical. Similarly, using US data spanning 1976–2009, Rogerson and Shimer (2011, pp. 624–625) find that over the business cycle, the labor force participation rate is nearly constant. Running a vector autoregression on US data for 1976–2016, Cairo, Fujita, and Morales-Jimenez (2022, figure 1C) find that the impulse response of the labor force participation rate to a positive productivity shock (the typical shock in the business cycle literature) is 0 for 2 years, and never significantly different from 0 after that.

It is true that labor force participation dropped around the Great Recession (Erceg and Levin 2014), but the decline was primarily caused by population aging and other trends that preceded the recession (Aaronson et al. 2014; Krueger 2017). The only recession that led to a big change in the participation rate was the pandemic recession. When the pandemic started, there was a large drop in the participation rate, from 63% down to 60%. This drop was something we had never seen before. Workers decided to drop out of the labor force in large numbers because it became dangerous to work, and school closures imposed new childcare obligations. The participation rate recovered once working became safe again and schools reopened.

2.6.3. Technology is broadly acyclical

Technology, which is a blueprint to effectively use capital and labor, is likely to evolve even more smoothly than the capital stock. Think about a restaurant: it is clearly easier to buy an additional microwave oven than design new technology for the microwave. Technology is therefore something that we take as fixed over the business cycle.

Although it is difficult to say for sure, technology is likely acyclical. The invention process is slow and random. The diffusion process is slow and random. The depreciation process—the loss of know-how—is also slow and random. We can look at some proxies for technology to convince ourselves. The number of transistors per microprocessor grows steadily, without cyclical fluctuations (Rupp 2023). The same is true of the number of operations that can be carried out per second by the fastest existing supercomputers (Dongarra, Luszczek, and Petit 2024).

It is true that technology is typically measured to be procyclical. But this is because technology is measured as a residual, whose fluctuations might be driven by slack (section 2.6.6).

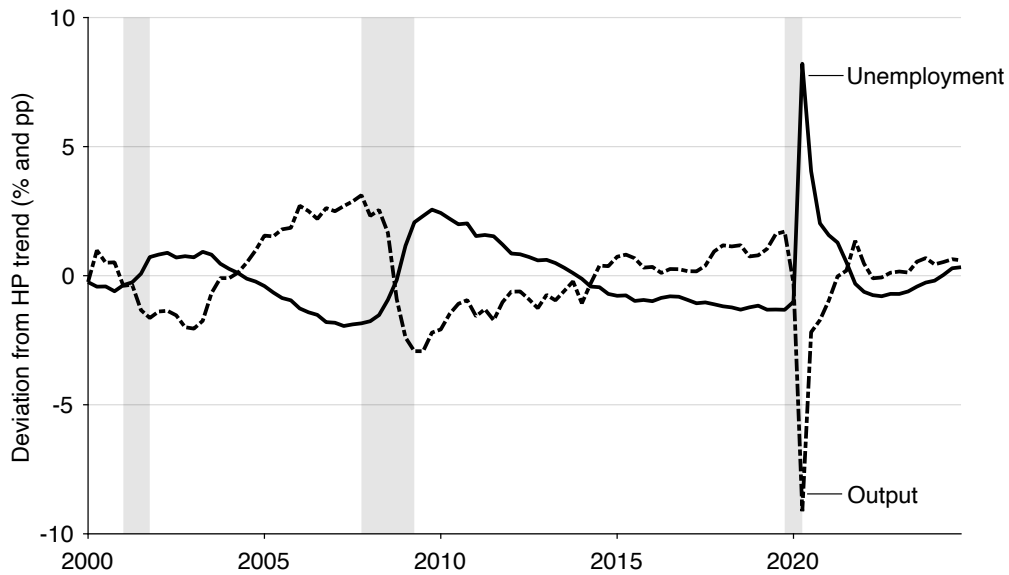
2.6.4. Slack is sharply countercyclical

We saw previously that slack on the labor and product markets was countercyclical: higher in bad times than in good times. We now examine whether such fluctuations could possibly explain fluctuations in output—which are the most common marker of business cycles. We focus on the last 25 years of data, from 2000:Q1 to 2024:Q4, both because slack data are only fully available since 2000, and because the link between slack and output became tighter in the 21st century.

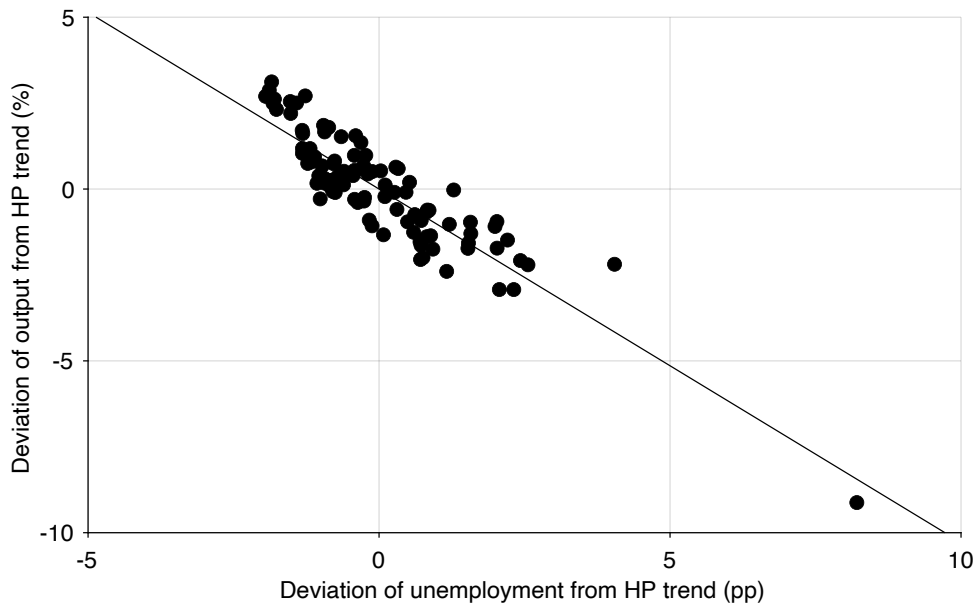
We first look at the connection between output and labor market slack. In figure 2.11A we superimpose cyclical fluctuations in output with those in unemployment. It appears that these fluctuations in unemployment could easily explain those in output since they are synchronized with them and of comparable amplitude. The correlation between the two series is almost -1 , at -0.90 .

In figure 2.11B we recast the results from figure 2.11A in the form of Okun (1963)'s law. Okun's law relates short-run movements in output and unemployment. In the United States, between 1948 and 2013, output and unemployment rate are negatively correlated in the short run: an increase in the unemployment rate by 1pp is associated with a reduction in output by about 2% (Ball, Leigh, and Loungani 2017). Between 2000 and 2024, the connection is instead one-for-one: a 1pp increase in unemployment is associated with a reduction in output by about 1%.

We also examine the connection between fluctuations in product market slack and output. Fluctuations in product market slack also contribute to fluctuations in output,



A. Correlation: -0.90



B. Estimated slope: -1.03

FIGURE 2.11. Okun's law with unemployment in the United States, 2000–2024

Detrended output comes from figure 2.10. Unemployment comes from figure 2.1. Unemployment is detrended by applying a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).

but because these slack data are noisier, the connection is not as tight as in the case of unemployment.

Figure 2.12 correlates cyclical fluctuations in output with those in nonmanufacturing idleness. The fluctuations are quite negatively correlated, with a correlation coefficient of -0.72 . So when there is more idle capacity in the nonmanufacturing sectors, output falls below trend. In Okun's law version with output and nonmanufacturing idleness, the slope is about $1/2$: an increase in nonmanufacturing idleness by 1pp is associated with a reduction in output by roughly 0.5%.

The link between output and idleness is weaker in the case of manufacturing (figure 2.13). Output does fall below trend when manufacturing idleness rises, but the correlation between the two series is only -0.65 . This is not so surprising, given that manufacturing now only employs less than 10% of the US workforce, and produces less than 15% of US GDP (Baily and Bosworth 2014, figure 1). In an Okun's law version with output and manufacturing idleness, the slope is $1/3$: an increase in the rate of manufacturing idleness by 1pp is associated with a reduction in output by about 0.3%.

2.6.5. Slack fluctuations are sufficient to account for output fluctuations

We have seen that in the US economy, a 1% change in output is associated with a 1pp change in unemployment rate (figure 2.11), a 2pp change in nonmanufacturing slack (figure 2.12), and a 3pp change in manufacturing slack (figure 2.13). We now show that these fluctuations in slack are more than sufficient to account for the fluctuations in output.

First, let's apply the slack identity (2.1) to the labor market. The slack identity becomes

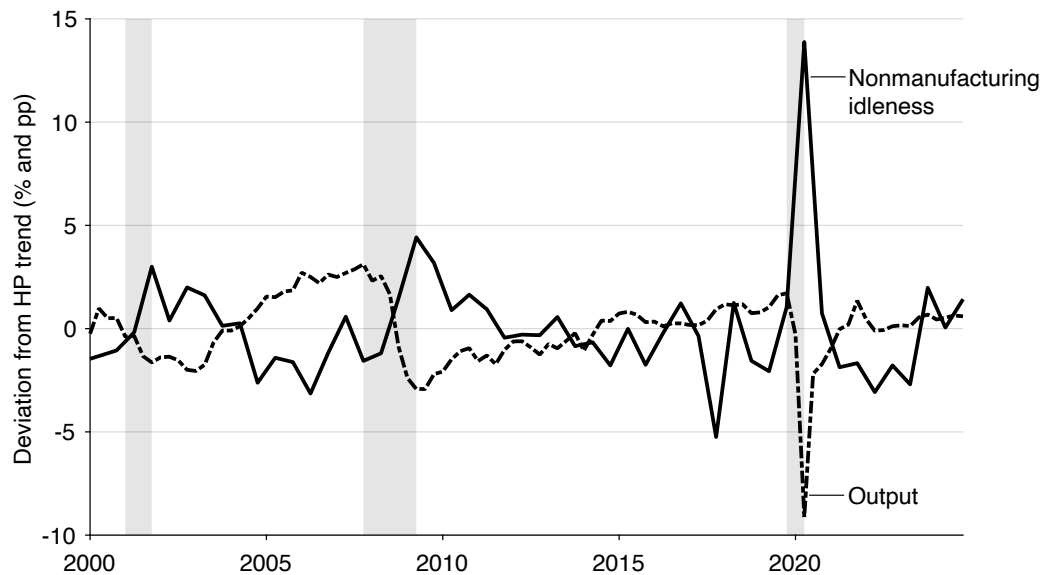
$$(2.2) \quad \text{labor} = (1 - \text{labor market slack}) \times \text{labor force}.$$

The main component of labor market slack is unemployment. When the unemployment rate increases by 1pp, the term $(1 - \text{labor market slack})$ falls by roughly 1%, since the unemployment rate is small compared to 1. So if the labor force is fixed, the amount of labor drops by roughly 1%.

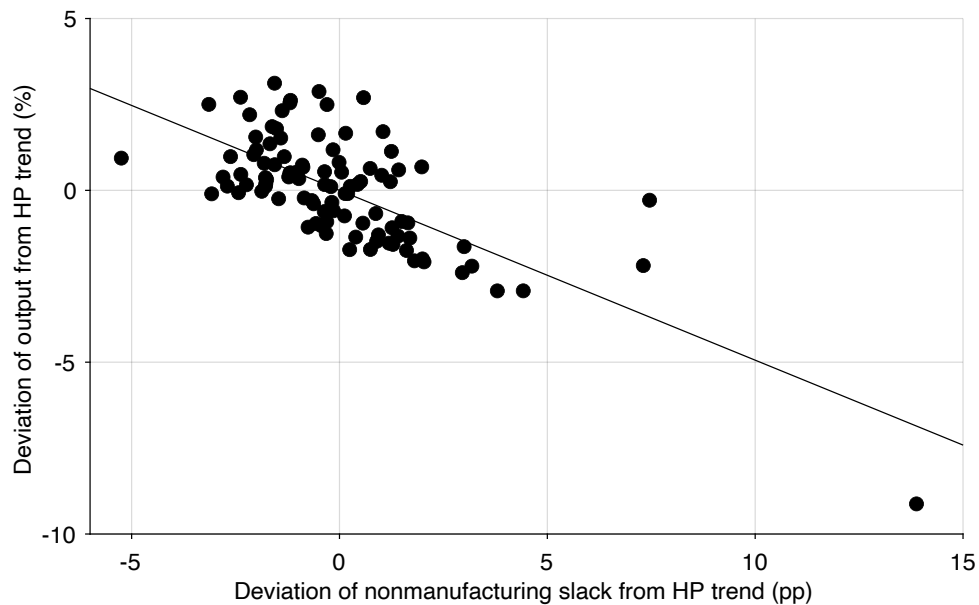
Next, we turn to the product market. The slack identity on the product market gives

$$(2.3) \quad \text{output} = (1 - \text{product market slack}) \times \text{capacity}(\text{technology, capital, labor}).$$

We use fluctuations in nonmanufacturing idleness as a measure of fluctuations in product market slack, since the nonmanufacturing sector is much larger than the manufacturing sector. We consider an increase in product market slack by 2pp, which again implies that the term $(1 - \text{product market slack})$ falls by roughly 2%. In addition, because labor falls by 1%, the productive capacity of firms, given by the production function, falls. Under



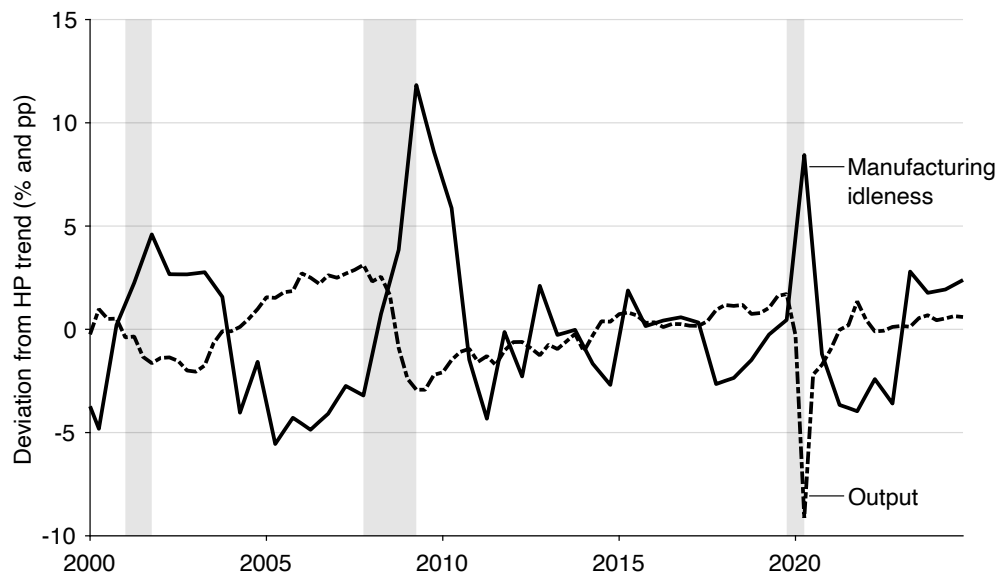
A. Correlation: -0.72



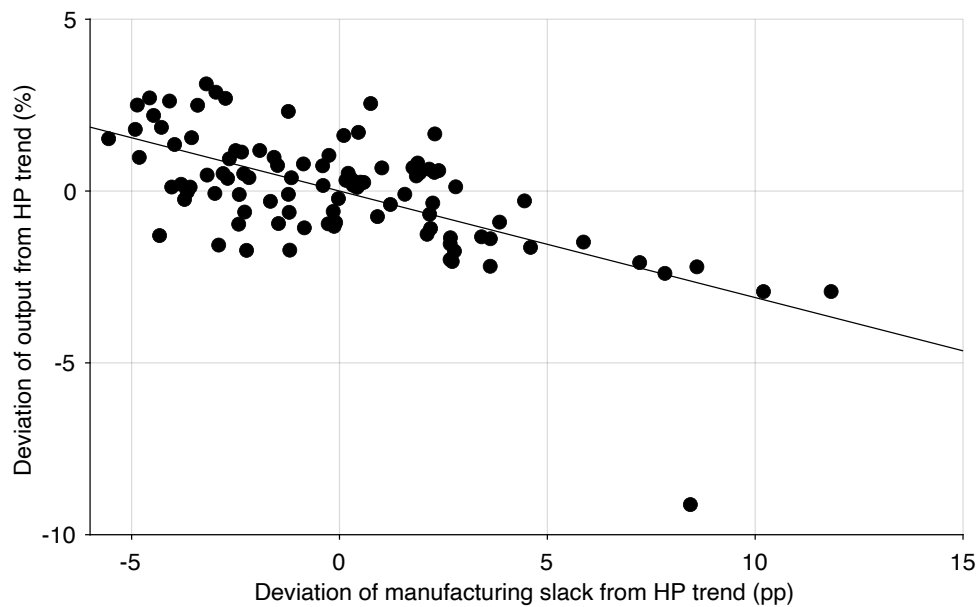
B. Estimated slope: -0.49

FIGURE 2.12. Okun's law with nonmanufacturing idleness in the United States, 2000–2024

Detrended output comes from figure 2.10. Nonmanufacturing idleness comes from figure 2.3. Idleness is detrended by applying a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).



A. Correlation: -0.65



B. Estimated slope: -0.31

FIGURE 2.13. Okun's law with manufacturing idleness in the United States, 2000–2024

Detrended output comes from figure 2.10. Manufacturing idleness comes from figure 2.3. Idleness is detrended by applying a HP filter with smoothing parameter of 10,000. Shaded areas indicate recessions dated by the NBER (2023).

the typical assumption that the aggregate production function takes a Cobb-Douglas form with an exponent $2/3$ on labor, then firms' productive capacity falls by $2/3\%$. Combining the drop in productive capacity with the increase in product market slack, we conclude that output decreases by $2\% + 2/3\% = 2.7\%$, which is more than the output drop of 1% that we seek to explain.

We conclude that the fluctuations in slack observed in the US economy can more than account for the fluctuations in US output, keeping all factors of production in the economy (labor force, capital stock, and technology) entirely fixed. Output fluctuations come from slack fluctuations on the labor market, through the mechanism described in (2.2), and from slack fluctuations on the product market, through the mechanism described in (2.3). In fact, we would expect even larger fluctuations in output given observed fluctuations in product market slack. A possible explanation is that the amount of idle capacity measured by the ISM (2025) is excessively volatile.

2.6.6. The illusion of procyclical technology

Furthermore, the framework suggests that procyclicality in measured technology, which is a cornerstone of modern business cycle models, may be an illusion created by unmeasured variations in slack. The evidence of the procyclicality of technology comes from measuring technology as a Solow residual. In the Solow procedure, output, capital, and labor are measured, but technology is inferred. The fluctuations in output that cannot be explained by fluctuations in capital and labor are residually assigned to fluctuations in technology.

In our slackish model, however, there can be fluctuations in output even if capital and labor are fixed, because the rate of utilization of capital and labor varies over time, as shown in equation (2.3). Thus, fluctuations in technology, measured as a residual, could very well be fluctuations in slack. If we are in a world with slack and perform the Solow procedure to measure technology, variations in utilization will be interpreted as variations in technology. For instance, if customers stop eating out in bad times, then restaurant workers will be idle more of the time, so measured technology—the amount of meals sold per worker—will decrease, although the restaurant technology has not changed. Fewer meals are sold not because the ovens stopped working but because fewer people showed up to dine in the restaurant.

Basu, Fernald, and Kimball (2006) propose a more sophisticated Solow procedure that attempts to remove the fluctuations in labor and capital utilization that pollute measured technology. Their purified measure of technology is much less variable than standard measured technology—as anticipated. In fact, accounting for variable utilization has such an impact that measured technology turns countercyclical—and can therefore in no way be a major driver of business cycles.⁶

⁶The countercyclicality of measured technology might be because good measures of slack are lacking.

2.6.7. Lack of progress in stabilizing slack

Before we enter the main part of the book, it is important to recognize the scope for policy: despite major progress in other areas, very little progress has been made in stabilizing slack and unemployment.

Politicians and policymakers traditionally worry about three issues: output and its growth, inflation, and unemployment. The US economy has experienced a tremendous amount of growth in the last decades of the 19th century and during the 20th century. Over the entire period, per-capita output has grown at an average rate of around 2% per year (Jones (2002, figure 1)). With such growth and finite human needs, it seems that the amount of output available for consumption is sufficient in most places. Consumption is not distributed equally across people, of course, because income and wealth remain unequal in the United States.⁷ Such inequality clearly remains a problem. But the total amount of consumption available seems sufficient.

Another common macroeconomic woe is inflation. The United States has struggled with inflation in the past: it was for instance a big issue in the 1970s. And inflation imposes noticeable social costs, primarily because people are angered by inflation.⁸ Since the 1990s, however, US inflation has been stable around low levels. For more than a quarter century, between 1994 and 2020, core inflation remained between 0.7% and 2.6%, despite tumultuous events that included the boom and bust caused by the dot-com bubble and the Great Recession (BEA 2025a). By contrast, in the previous quarter century, between 1969 and 1993, core inflation varied between 2.5% and 10.1%. So the Federal Reserve has made tremendous progress in taming inflation. And such progress has undoubtedly been enabled by the insights from the New Keynesian literature. Obviously, US inflation has reached excessive levels in the aftermath of the coronavirus pandemic, but it subsided almost entirely within about 3 years. Time will tell if inflation returns more permanently.

Unlike other macroeconomic problems, little progress has been made about unemployment. The problem of unemployment manifests itself in several ways. First, over time, the average level of unemployment has not noticeably fallen (figure 2.1). Moreover, the peaks of unemployment continue to reach high levels. Of course, unemployment has never reached again the levels from the Great Depression—but the pandemic recession saw the highest peak in unemployment since 1940, and the Great Recession the third highest. Last, for certain socioeconomic groups, unemployment remains stubbornly high (Cairo and Cajner 2018, figure 1).

Basu, Fernald, and Kimball (2006) do observe hours per worker, which is one form of product market slack, as discussed in section 2.3.3. But they do not observe idle labor or idle capital in firms. Instead, they must input them by assuming that underutilization of labor and capital is voluntary, to minimize costs. In a world with slack, underutilization of labor and capital is involuntary, due to insufficient demand. Reproducing their procedure with direct measures of idle labor and capital might well yield acyclical technology.

⁷ See for instance Piketty and Saez (2003), Saez and Zucman (2016), and Piketty, Saez, and Zucman (2018).

⁸ See for instance Okun (1981, pp. 278–289), Shiller (1997), Stantcheva (2024), and Georgarakos et al. (2025).

This lack of progress is likely due to the lack of slack in modern business cycle models: slack has been forgotten in the Real Business Cycle and New Keynesian models. That is why slack is the focus of this book. With a framework centered on slack, hopefully, policy will improve and fluctuations in unemployment and other forms of slack will subside, just like fluctuations in inflation have.

2.7. Social cost of slack

Finally, before building a theory of economic slack, we establish why having slack in a society is so problematic. In the first place, slack is costly because it represents a waste of productive resources. Labor that is available to produce value is left unemployed, and goods and services that are available to be consumed are left unsold. Additionally, the idleness associated with unemployment creates psychological hardship. A final cost arises from the matching efforts exerted by buyers in a slackish world. Recruiting on the labor market and shopping on the product market absorb time and labor which could otherwise be used to produce valuable goods and services.

2.7.1. Social cost of labor market slack

Labor market slack takes the form of unemployed or underemployed labor. Mechanically, this is costly because it is a waste of productive resources. Labor that is available to generate value once incorporated in firms' production process is left unused.

Now, in economics, we usually assume that leisure is valuable—this explains why people do not work all the time. Additional leisure time could in principle alleviate the social cost of unemployment. But, as Bakke (1969, p. 16) concludes after observing and interviewing unemployed workers for several years in New Haven during the Great Depression, “unemployment is not leisure.” Bakke (1969, table 1) shows that unemployed workers engage in much fewer recreational activities than employed workers, despite having much more time on their hands. Among 24 recreational activities, they do strictly less of 18 activities, and strictly more of only 5 activities: sitting around home, sleeping during daytime, chatting and gossiping at corner stores and on the street, walking, and gardening and home repairs. So, of the activities that unemployed workers do more often than employed workers, 4 are various forms of being idle. As Bakke (1969, p. 16) notes, “Several hours in every day have been transferred to the leisure account. But the transfer is made under circumstances that cancel many of the opportunities usually related to leisure.” One reason that he isolates is that workers who lose their jobs also lose their place in society, and their social network, which cuts them off from many forms of recreation. A second reason is that it is difficult for workers to change their routine and take part in new recreational activities. As Bakke (1969, p. 17) writes: “The plain fact is that these

[activities] are not a normal part of their life; they have little or no experience with them; they have no normal incentive to make use of them. ”

In fact, being unemployed is an extremely traumatic experience—among the worst events that can happen to an adult. That is why, beyond the wastefulness that it produces, unemployment generates large additional costs to society. People who are unemployed tend to suffer from lower mental health and lower physical health than people who are employed. Robinson (1949, p. 11) for instance observed that “The most striking aspect of unemployment is the suffering of the unemployed and their families—the loss of health and morale that follows loss of income and occupation.” At this point, the detrimental effects of unemployment on mental, physical, and public health are well documented.⁹

Critically, unemployment carries substantial costs even when controlling for income and personal characteristics. In the US General Social Survey, Di Tella, MacCulloch, and Oswald (2003) find that the nonpecuniary cost from becoming unemployed can be as profound as the emotional distress experienced when going through a divorce or dropping from the top income quartile to the bottom. In the same survey, Blanchflower and Oswald (2004, p. 1373) confirm that the nonpecuniary toll of unemployment can be substantial, akin to the emotional impact of losing \$60,000 in annual income. Using two large, daily US surveys, Helliwell and Huang (2014) confirm that for workers who have fallen into unemployment, the nonpecuniary costs are several times as large as the pecuniary costs. They also find that an increase in local unemployment reduces the well-being of workers who are still employed much more than what the reduction in local income would predict.

Where do the nonpecuniary costs of unemployment come from? There are various sources, which became clearly visible as early as the Great Depression. Eisenberg and Lazarsfeld (1938) review findings from clinicians who worked with unemployed workers, in the field, during the Great Depression. They isolate several causes for the elevated anxiety and widespread depression among jobless workers:

Unemployment represents a personal threat to an individual’s economic security; fear plays a large role; . . . the individual’s prestige is lost in his own eyes, and as he imagines, in the eyes of his fellow men. He develops feelings of inferiority, loses his self-confidence, and in general, loses his morale.

What’s fascinating is that all these negative facets of unemployment have been confirmed since. Bakke (1969) finds that unemployment strain family and other social relations. To illustrate the strain on personal relationships, and the social stigma associated with joblessness, Bakke (1969, p. 12) quotes a carpenter who was unemployed during the Great Depression:

⁹See for example Dooley, Fielding, and Levi (1996), Platt and Hawton (2000), Frey and Stutzer (2002), McKee-Ryan et al. (2005), Wanberg (2012), and Brand (2015).

Every time I go to the home of one of my friends, they will say, ‘Well, how about it? Do you have a job yet?’ and constantly I’m being reminded of the fact that I am out of work. I hate to go even to see my relatives because I know that they’re thinking ‘So-and-so has a job; why can’t Jim get one?’ Perhaps this is only all in my own mind, but it keeps me from enjoying these relationships with them just the same.

Goldsmith and Darity (1992) provide a trove of evidence from psychology showing that an important reason why joblessness diminishes well-being is that it creates a sense of helplessness: that one’s life is no longer under their control. Goldsmith, Veum, and William Jr. (1996) offer evidence that job loss and unemployment lead to a decline in an individual’s self-esteem and sense of self-worth. In addition, Krueger and Mueller (2011) find that the process of unsuccessfully searching for a job significantly reduces unemployed workers’ life satisfaction.

In fact, Jahoda (1981) emphasizes numerous important benefits of work—which are lost during unemployment. A lot of people care about their job and derive a lot of meaning from their employment. Their personal status and identity come from their jobs. Other benefits from work encompass a structured daily routine, regular interactions and shared experiences and goals with individuals beyond the immediate family, and the engagement in regular activities. Collectively, the loss of these benefits contributes to the psychological burdens associated with unemployment.

Overall, then, unemployed workers divide their time on three possible activities. The first is looking for a job. The second is producing goods and services at home. The third is remaining idle—maybe waiting for call backs from potential employers. How do these three activities contribute to social output? Jobseeking is required to find employment but—just like recruiting—does not contribute to social output. Home production adds to social output, while the psychological cost of idleness depletes social output.

As a final step, we assess the social product of unemployed labor. The value of job seekers’ home production, net of the psychological cost of idleness, is estimated to be negligible. Using administrative data from the US military, Borgschulte and Martorell (2018) study how servicemembers choose between reenlisting and leaving the military. The choices allow them to estimate the value of home production plus public benefits received during unemployment minus the psychological cost of idleness. Subtracting the value of public benefits from these estimates, Michailat and Saez (2021, p. 11) find that the value of home production minus the psychological cost of idleness could be as low as 3% of the value of market production. Given its minimal value, we take the social product of unemployed labor to be zero.¹⁰

¹⁰Okun (1981, pp. 272–274) notes that unemployed workers have time available “for seeking other jobs” and “for the variety of pursuits that may be described as home activity”. Then, to assess the social costs

2.7.2. Social cost of product market slack

All goods, nondurable and durable, depreciate when they are not sold, which is socially costly. The most costly such form of slack might be unsold food—both because food is the most necessary of goods, and because food perishes quickly.

Services that are not sold are obviously immediately lost. So the idle capacity in non-manufacturing firms documented in figure 2.3 imposes a direct welfare cost, since this capacity was readily available to provide value to customers.

Another cost of product market slack takes the form of advertising. Because firms find it difficult to sell their production, they divert resources to advertising and selling it, in the hope of improving sales (Hall 2012). The OES indicates that 11% of US workers are in occupations devoted to sales and advertising, including retail salespersons, sales representatives, and advertising agents (Gourio and Rudanko 2014). Arkolakis (2010, Appendix A) reports that 4%–8% of US GDP is devoted to marketing, with advertising alone constituting 2%–3% of GDP. Relatedly, based on a survey of 4,000 firms run by APQC, Fernandez-Villaverde et al. (2025) report that 7.5% of firms' revenue is devoted to attracting customers. From a social perspective, the activity of sales, marketing, and advertising are wasteful: they are required for sellers to find buyers, but do not directly produce welfare for consumers.

2.7.3. Social cost of recruiting

Employed workers must spend some of their time recruiting new hires for their firms, so they are unable to spend their entire time contributing to social output. Recruiting takes work: designing and advertising job vacancies, screening and interviewing candidates, and negotiating contracts.

Unfortunately, so far, there is only limited evidence on the number of recruiters in firms or the number of man-hours devoted to recruiting by firms. There are two sources of information about the amount of labor devoted to recruiting in the United States.

The first source of information about the amount of labor devoted to recruiting in the United States is from the National Employer Survey, which was conducted by the Census Bureau in 1997 (Villena Roldan 2010). The survey asked thousands of establishments across industries about their recruiting practices (Cappelli 2001). On average, establishments report that they devote 3.2% of labor costs to recruiting, which implies that servicing a job vacancy requires 0.92 workers at any point in time (Michaillat and Saez 2021, p. 11).

A second source is the survey conducted by the consulting firm Bersin and Associates in 2011 (Gavazza, Mongey, and Violante 2018). The survey asked over 400 firms with more

of recessions and slack, he weighs the value of job seekers' home activity against "the social stigma and family strains associated with unemployment." He agrees with us that the value of home activity net of the psychological cost of unemployment is negligible.

than 100 employees about their spending on all recruiting activities. Firms reported that recruiting one worker costs 93% of a monthly wage, which implies that it takes 1.16 workers to service a vacancy at any point in time (Michaillat and Saez 2024, p. 332).

Both surveys show that in the United States, it takes about 1 full-time worker to service a job vacancy. Thus, the number of vacancies is a good measure of the number of recruiters in the economy: the number of workers diverted from producing and allocated to recruiting can be measured by the number of vacancies.

2.7.4. Social costs of shopping

A first social cost of shopping comes from sheer amount of time spent by consumers shopping. Time-use data recorded by the BLS through the American Time Use Survey (ATUS) show that on an average day between 2003 and 2012, US consumers devote 42 minutes to shopping, including 18 minutes traveling to shops, 8 minutes buying groceries, gas, and food, 15 minutes researching and buying consumer goods and services, and 1 minute waiting at shops (Petrosky-Nadeau, Wasmer, and Zeng 2016, table 1).

A second social cost of shopping arises from the vast amount of goods that are returned to retailers. We saw earlier that 10% of all retail sales, and 18% of online retail sales, are returned to sellers because the goods do not fulfill the needs of the buyers. These returns have substantial social costs, including extra shipping but especially the depreciation of the returned good. Indeed, returned goods can only rarely be sold as new; they are often sold at a discount or even discarded (Reagan 2019).

Shoppers also face stockouts, which are visibly costly: buyers are generally upset when they drive to a shop, expecting to buy a product, but are unable to find it (Taylor and Fawcett 2001). In a field experiment conducted in a mail-order catalog, Anderson, Fitzsimons, and Simester (2006) find that customers subjected to a stockout reduce their spending on other items in the order, and reduce long-run spending on the catalog's products. As Okun (1981, p. 276) notes, shortages and stockouts push buyers to "react in ways that involve costs—ordering further in advance, making commitments that sacrifice flexibility, and the like." Some of the goods that are bought in advance for fear of stockouts are eventually not needed and wasted.

Finally, firms devote labor both to buying goods and services. As such labor is diverted from producing goods and services, it imposes a social cost. In data recorded by the OES between 2003 and 2021, firms allocate 1.4% of employment to ordering, buying, purchasing, and procurement (Fernandez-Villaverde et al. 2025, p. 2508).

2.8. Summary

In this chapter we saw that economic slack—the prevalence of unsold goods and services—is a pervasive feature of the US economy, not just a phenomenon of recessions. On the labor market, unemployment is always present, averaging about 6% since 1929 and spiking during recessions. On the product market, slack is even more prevalent. On average, firms operate with 14%–17% of capacity that is idle. Because slack is so prevalent, the Walrasian model is not well-suited to describe real-world markets.

An empirical puzzle is the coexistence of sellers and buyers who cannot find each other. In the labor market, this is captured by the Beveridge curve, which shows a negative relationship between unemployed workers and vacant jobs. It's because of this coexistence of vacant jobs and unemployed workers that we cannot use the nonclearing Walrasian model and must develop a new theory of slack. In the product market, we also always have buyers trying to purchase goods and services, together with goods and services for sale. This coexistence of buyers and sellers on markets points to a fundamental matching problem in the economy.

The chapter argues that business cycles correspond to fluctuations in slack, not in productive capacity. Inputs like capital stock, labor force, and technology are largely acyclical. In contrast, labor and product market slack is sharply countercyclical. Fluctuations in slack are more than sufficient to account for output fluctuations.

Finally, the chapter argues that slack is socially costly. First, it wastes productive resources: goods and services that could generate value remain unsold. Second, on the labor market, unemployment imposes severe nonpecuniary costs, because many people derive identity, status, and daily structure from work. Last, the presence of slack is associated with matching efforts—such as recruiting and shopping—that absorb resources which could otherwise be used to produce goods and services.

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